Signal processing: methods and algorithms
Laboratory 2

1 Time-varying random systems

This exercise requires the generation of nonstationary random processes modeled by

$$X[n] = \sigma[n]F[n] + \mu[n]$$

where $F[n]$ is a discrete-time white Gaussian noise with zero mean and unit variance, whereas $\mu[n]$ and $\sigma[n]$ are deterministic functions. Do the following:

1. Generate and plot $M = 5$ realizations of $F[n]$. Plot the realizations on the same figure, by using the `subplot` function.
2. Take $\sigma[n] = 1$ and, by changing the function $\mu[n]$, generate a random process $X[n]$ with time-varying mean. The function $\mu[n]$ must have the following behaviors:

   2.1 *Linear:* $\mu[n]$ changes linearly from μ_1 to μ_2, where μ_1 and μ_2 are arbitrary values.

   2.2 *Piecewise constant:* $\mu[n]$ is constant over an interval, but changes from one interval to the other. Choose the number and duration of the intervals at will, as well as the value of $\mu[n]$ over each interval.

   2.3 *Cyclic:* Let $\mu[n]$ be a sinusoidal function, that oscillates between two arbitrary values μ_1 and μ_2.

3. Take $\mu[n] = 0$ and, by changing $\sigma[n]$, generate a random process with time-varying variance. The function $\sigma[n]$ must have the following behaviors:

   3.1 *Linear:* $\sigma[n]$ changes linearly from σ_1 to σ_2, where σ_1 and σ_2 are arbitrary values.

   3.2 *Piecewise constant:* $\sigma[n]$ is constant over an interval, but changes from one interval to another. Choose the number and duration of the intervals at will, as well as the value of $\sigma[n]$ over each interval.

   3.3 *Cyclic:* Let $\sigma[n]$ be a sinusoidal function that oscillates between two arbitrary values σ_1 and σ_2.